MERIT-ML User Manual

Evaluating ML readiness in tabular metabolomics datasets from Metabolomics Workbench

Manual version: **12 November 2026**. This guide describes the current browser and native Docker applications. It supersedes the v7 manual and its composite scores, readiness bands and feasibility gates.

MERIT-ML means *Metabolomics Evaluation of Readiness and Interoperability of Tabular Data for Machine Learning*. Open MERIT-ML to assess a public Metabolomics Workbench study, or use the Docker application and source code on your own computer. MERIT-ML is independent of Metabolomics Workbench.

This guide includes every current scored item, the separate Structural and FAIR summaries, informational blocks, all 13 numerical settings, the eight custom alert cut-offs, class selection, sample and subject edits, guidance, downloads, bulk use and resource-limit messages. The software controls, rather than historical manuscript figures, define the current interface.

Contents

1. Start here
2. Studies, analyses, data files and denominators
3. Read the Overview
4. Choose the classification task
5. Adjust samples, labels, batches and subjects
6. Structural diagnostics
7. Metadata and FAIR Reusability
8. Analytical QC: samples, controls and processing
9. Analytical QC: the four alerted metrics
10. Analytical QC: scale, units and contextual evidence
11. Annotation / Interoperability
12. Label Structure and Class Support metrics
13. ML Task Readiness metrics
14. All numerical controls
15. R1 and alert interpretation
16. Downloads and provenance
17. Bulk assessment and local Docker use
18. Browser limits and partial results
19. Troubleshooting and practical interpretation
20. Complete metric and evidence checklist
21. Versions, provenance and further help

1. Start here

1. Enter a public Workbench study accession, for example `ST000001`, and start the assessment. A recent-study card can also fill the search and start that study.
2. Wait for input retrieval and the named calculation steps. **Generating report** follows the calculations; reaching the end of the calculation progress bar is not yet the finished report.
3. Read **Data at a glance** and the **Readiness profile** on Overview. Check each analysis and data file; the sources may disagree.
4. Define the classification task using the class editor on Overview or in **Label Structure and Class Support**. Select the number of cross-validation folds in **Show tools** and apply the change.
5. Review sample eligibility, repeated sample IDs, subject IDs, controls, analytical alerts and checks that could not be assessed. Change inputs only when you can justify the change from the deposit or your scientific question.
6. Inspect the metric tabs and save the assessment JSON, relevant feature lists and matrices. Preprocessing, feature selection and model fitting remain your responsibility.

The browser assesses one study at a time using your computer. Workbench response time, network speed, matrix size and available memory affect duration. The Docker application also retrieves inputs on demand and supports bulk assessment. Neither needs the frozen manuscript study-result cache to calculate a new assessment. Both use versioned reference assets, including R1 and chemical lookup data; those are distinct from precomputed study assessments.

1.1 Search, availability and appearance

Use the study's `ST` accession, not a project, analysis (`AN`) or metabolite identifier. The application checks public availability. An invalid accession, an unavailable study, an explicitly identified embargo and a failed availability request are different states. A metadata-only report can be valid when no readable quantitative matrix is supplied through the supported endpoints.

About, **Publication**, **GitHub** and **User manual** provide scope, methods, source/installation instructions and this guide. The light/dark toggle changes display only. Searching for another study starts a separate assessment; it must not reuse the previous study's scientific results.

Recent studies are discovery aids, not a readiness ranking. If discovery cannot be loaded, use the accession search or the Workbench link. A raw-data file or an Excel attachment elsewhere in a study does not guarantee that one of the supported quantitative endpoints contains a readable matrix. MERIT-ML does not reconstruct feature tables from raw spectra or chromatograms.

1.2 Guided tours

A search-page tour and a report tour explain the interface separately. Use **Next**, **Back**, **Skip tour**, the close button or Escape. **Don't show tours automatically** opts out; **Take a tour** replays the relevant tour when you want it.

Tour state is saved for that browser and site, so opening another study or applying settings does not repeatedly launch it. Different browsers, private sessions, host ports and cleared site storage can have separate preferences. When persistent browser storage is blocked, session storage or the open page's memory is used instead. Tour preferences are not scientific inputs and are not sent as sample data. The tour can open existing tabs or details for explanation; it does not change classes, confirm subjects, alter parameters or submit an assessment.

2. Studies, analyses, data files and denominators

2.1 What one result represents

A study can contain several analyses. Each analysis can contain independently readable **datatable**, **mwTab** and **untarg_data** matrices. A **matrix** in this guide means one analysis × endpoint. The readiness profile never averages endpoints or analyses into one study score.

Term	Meaning
datatable	Quantitative table obtained from the official `/datatable/` endpoint, not `/datatable/file/`.
mwTab	A readable quantitative metabolite-data block in the analysis's mwTab record. The record can contain metadata without a valid numeric matrix.
untarg_data	A readable untargeted feature table from the official untarg_data endpoint. Registry membership alone does not establish matrix validity.
Tier 1	The manuscript's grouping of datatable and mwTab. They remain separate sources and can differ in samples, features and values.
Tier 2	The manuscript's grouping of untarg_data. The tier is a source convention, not an analytical-quality ranking.
Position	One sample row of datatable or untarg_data, or one sample column of an mwTab quantitative block.
Distinct sample ID	A sample identifier counted once even when it occurs at several positions. It does not necessarily identify an independent subject.
ML-eligible position	A measured position remaining after the current eligibility rules and edits, excluding Tier 1 alignment rows and controls not selected for modelling.
Selected-class sample	An ML-eligible sample ID assigned to one of the classes currently selected. Not every eligible sample has to belong to those classes.

Different metrics use different denominators deliberately. Cell and sample-median diagnostics generally use **ML-eligible positions**; class support and separability use **distinct sample IDs in the classes in force**. A feature denominator can be all feature positions, judged features, evaluated non-low-information features, eligible feature labels or named features. These are specified for each metric below. Do not interchange them.

2.2 Tier 1 sample alignment

A datatable can contain an additional all-zero sample row whose sample ID is absent from the paired mwTab quantitative block. MERIT-ML identifies the specific Tier 1 alignment condition, reports the affected counts and IDs, and excludes those positions from analytical and class calculations. It does not establish from this alone whether a physical sample was ever experimentally measured.

An alignment position cannot be made ML-eligible by an edit. The same sample ID may nevertheless be measured in another analysis; the sample editor's per-matrix status strip shows that difference. Do not delete every all-zero row indiscriminately: an all-zero row with a measured counterpart is a different case.

2.3 Controls and missing identifiers

Controls are recognised as **QC-like**, **blank-like** or **standard-like**, not experimentally verified. Default recognition considers sample IDs and factor values, sample-table evidence and the distributed curated corrections. Details identify the keyword or correction and where it came from. Unrecognised controls can remain eligible until you review them.

Empty factor fields are distinct from deposited words such as `NA`, `None` or `Unknown`. Those words can be genuine levels, for example an untreated control, so class definitions do not automatically delete them. Subject-placeholder handling treats empty, `-`, NA, N/A, NaN, null, none, unknown and not applicable (case-insensitive) as no subject. Numeric-cell missing-token handling has its own rules; a string's meaning is not assumed to be identical in every context.

3. Read the Overview

3.1 Study information and data-source availability

The report begins **MERIT-ML Assessment Report — ST…** and identifies the study and its available analyses, quantitative endpoints, repository metadata and citation. Disease, organism, sample source and other descriptive fields are repository evidence, not inferred prediction targets. For example, the disease display uses the dedicated Workbench disease evidence; a displayed disease does not automatically define the selected classes.

The data-source information distinguishes present valid matrices, records without a readable matrix, unavailable endpoints and failures. In the browser, a resource refusal is shown separately from absence. If one endpoint cannot be assessed, other available endpoints and analyses can still produce results. Follow the named endpoint and omission reason rather than interpreting a blank as absence or zero.

3.2 Data at a glance

This collapsible block gives a deterministic plain-language explanation, not a generated diagnosis or a new score. It describes mwTab when available (otherwise datatable), and untarg_data separately. The full profile below still includes every assessed endpoint.

Its analysis cards explain, where applicable:

- measured positions, distinct ML-eligible sample IDs, controls, Tier 1 alignment and positions set aside by edits;
- feature counts and the annotation classes actually found, such as named metabolites, m/z–RT features or NMR bins;
- Structural items and explicit reasons for unavailable items;
- the active classes, class sizes, selected samples, count requirement, class support, separability and its chance level;
- missing and zero categories, low-information features, redundancy and sample-median outliers, with the supporting burdens and active settings;
- QC reproducibility, unit/scale cautions, subject sharing, batch-like evidence and important restrictions where available;
- task-dependent feature-association results, actual test names or the reason no test ran;
- practical review guidance and source-matrix downloads.

The narrative follows **applied** settings, classes and edits. A control you have changed but not applied is not yet the active assessment. No eligible samples, no samples in selected classes, unavailable checks and calculations that could not be recomputed are stated explicitly. The text must not be read as four passing checks when only one was assessed.

3.3 Derived context

Study-design context summarises evidence for cohort/subject-based, experimental/replicate-based or unclear design. It is a MERIT-derived interpretation, not a native repository field or a readiness gate. Confidence and source evidence help interpret small sample counts; a cell-culture experiment need not have the same aim as a population classifier.

Analyzer capability and acquisition evidence separate instrument-derived HRMS-capable, nominal/unit-resolution and indeterminate classifications from acquisition wording. Analyzer capability does not prove the resolving power achieved, targeted/untargeted status, annotation quality or experimental quality. Targeted/untargeted-like wording, scan-mode terms, an untargeted-file marker and analysis-linked MS2 evidence are independent evidence streams, with their scope and provenance. Evidence in a study-wide section need not describe each analysis. Historical contextual reference evidence, where used, is labelled separately from fresh programmatic evidence.

3.4 Readiness-profile columns

Column	What to read
Analysis	The AN accession; keep different analyses separate.
Data file	datatable, mwTab or untarg_data. Grouped endpoints have the same summary signature; their numeric values can still differ.
Summary	Count-requirement status, strongest analytical alert and assessed coverage, with relevant subject/structural notes. It is not a readiness band.
Cross-validation sample-count support	Smallest class, required folds, class/sample counts, p/n, class support, separability/chance and subject warnings.
Analytical alerts · fixed reference	The named checks raising low/extreme alerts, their values/cut-offs, checks unavailable and units documentation.
Selected annotation and metadata indicators	Named-feature coverage, named features mapped to RefMet, external IDs, mass/RT-field presence and LC-method context. These do not set the alert/count summary.

Count requirement met means at least two classes and at least `k` distinct labelled eligible sample IDs in the smallest class. **Count requirement not met** includes one class or an insufficient smallest class. **Cannot assess class counts** includes no labelled eligible samples or an absent selected class in that particular matrix. An unreadable or missing matrix is a separate availability state.

No alerts means no assessed check crosses the current low/extreme boundary. **Low** means at least one low alert and no extreme alert. **Extreme** means at least one extreme alert, potentially with additional low alerts. The numbers of each are shown. **Not assessed** names a missing calculation, not a successful check.

Read `4/4 checks assessed` independently of the alert label. With fewer than four assessed, a no-alert result is incomplete evidence. All four can be assessed and all four can raise alerts; the summary uses the strongest tier and reports the counts, rather than an average. Units declared is outside those four.

“Same summary” does not mean byte-identical matrices or identical annotation, feature counts or p/n. Differences are retained within the grouped row or in metric tabs. No endpoint is selected as a universal winner.

3.5 Definitions, tiles, colour and supporting evidence

Expand the readiness description and cut-off panel for the active reference, settings and absolute burdens. The **Scores · every analysis × endpoint × item** grid preserves individual values. Use **Show numbers**, the legend, hover/focus/tap help and section links; colours alone are insufficient. Hatched/unavailable/no-matrix states differ from numerical zero.

Individual metric headings explain what a higher value means. A high annotation score means more name/mapping coverage; it is not proof that classes can be predicted. A small RSD is generally preferable for reproducibility even though RSD is informational. Units, instrument categories and numerical scale must be interpreted in their own context. The Structural mean and FAIR documentation index are separate diagnostic summaries; neither contributes to the readiness-profile decision.

4. Choose the classification task

4.1 Whole strings, factors and observed combinations

By default each distinct complete `SUBJECT_SAMPLE_FACTORS` string is a class. For a synthetic label `Treatment: control | Time: 4 h | Sex: female`, the whole three-factor combination is one default class. Selecting Treatment alone pools every selected sample with the same Treatment value across Time and Sex; it does not hold those factors constant.

MERIT splits a label conservatively at the spaced delimiter `|` : every segment must have a unique non-empty factor name followed by `:`. It does not split an embedded unspaced pipe inside a value. Unreadable, ambiguous or conflicting labels are reported rather than silently assigned to an invented class.

The selector uses the study-wide available factors and observed value combinations, assembled across its matrices. A factor needs at least two values to be selectable. Choose one or several factors and at least two **observed** classes. A class can be a single value or a joint combination; the selector does not generate a Cartesian product of combinations that never occurred. Selecting several values does not merge them into one class.

Study-wide option counts are discovery counts. They do not guarantee that a combination is measured, eligible or present in every matrix. After **Apply classes**, each matrix is counted from its own eligible sample IDs. A factor that varies in the study can be constant in one matrix; a chosen class absent in that matrix makes its class-dependent assessment unavailable. A factor may disappear or a class choice may reset after edits; the report states a choice that could not be applied.

Use **Use the whole factor string as deposited** to restore default classes. Overview and Label Structure share the same class definition. Selecting another scientifically meaningful task can change count support, class support, separability, feature associations and task-specific recommendations; it does not create samples or improve the original measurements.

4.2 Samples left out and denominator checks

The tab reports the class definition, every class count, smallest/largest class and samples outside the chosen task. Reasons include no label, conflicting labels, another value/combination, an unsplittable string or a missing selected factor. Invalid study-wide choices fall back with an explanation; a valid study-wide choice with a zero-sized class in one matrix is not silently reduced to fewer classes there.

Repeated sample IDs count once for class support. Contradictory sample-table labels for one sample ID leave that ID unassigned unless corrected. Empty labels are separate from an explicit word such as `NA`. Choosing a label is a scientific decision; do not select a task only to obtain a met condition or stronger separability.

4.3 Other factors across the chosen classes

This informational block appears for a selected-factor task. Each remaining deposited factor is compared on the selected sample IDs. The first applicable state is reported:

State	Exact interpretation
Not assessable	Some selected samples do not report the factor; coverage is shown.
One reported level	Every selected sample has the same reported value.
One value per sample	Every value belongs to just one selected sample, so levels are not shared.
Values absent from some classes	At least one observed value occurs in one selected class but has zero observations in another. This is an observed absence, not a missing-field count.
Same proportions	Every value has exactly the same share in every selected class, compared using counts rather than rounded percentages.
Different proportions	Every value occurs in every class, but shares differ. The largest gap is shown in percentage points.

Qualifying remaining factors are also compared jointly when at least two are varying, fully reported and shared between samples. Details identify the factors and counts; value sets above 20 levels can omit the small hover count table. There is no imbalance score or statistical significance threshold. Equal observed proportions do not establish freedom from confounding: undeclared factors, subjects, batch, run order and interactions outside the recorded factors can remain important.

5. Adjust samples, labels, batches and subjects

Open **Adjust Matrix Properties**. There is one row per sample ID of the study; an edit applies wherever that ID occurs, not just to one selected endpoint. Matrix-status strips show which analyses/endpoints contain it, whether it occupies several positions, its measured/alignment/control/eligible status and which statuses changed.

Control	Effect after Apply sample edits
Use as: as deposited	Restore default eligibility for that ID.
Use as: ML-eligible	Include measured positions for modelling where allowed. Tier 1 alignment positions remain ineligible.
Use as: not ML-eligible	Set aside the measured sample without claiming it is a QC or blank.
Use as: leave out	Exclude the sample by an explicit edit; this is reported separately from controls.
Factor label	Replace the label used for that sample; blank edit fields retain the deposited/default label.
Batch-like label	Override the label used by Sample annotations; `-` gives it no batch-like label. This is informational.
Subject	Override the deposited subject; `-` gives it no subject. Blank retains the deposited value.
Subject confirmation	Confirm that the subject values identify real persons, animals or sources, not group names or placeholders. This enables supported subject-aware feature tests.

The **Show** filters select visible statuses; grouping only reorganises the table. The bulk **set all shown rows to…** control affects rows currently visible, then requires Apply. Subject shortcuts set every listed row to its own subject, no subject or the deposited subjects; Apply is still required. Inspect their scope before using them. Assigning one subject per sample is an edit, not evidence that those samples are independent.

Deposited evidence shows sample/subject IDs, factor values, the keyword and source behind control status or a curated decision, and additional sample data. A generic shared subject such as a species or “Human” can be an unsuitable subject identifier. Confirm only after checking what the value means. Missing subject IDs do not establish independence.

Apply sample edits recalculates applicable panels and downloads; **Clear all edits** restores deposited/default inputs. None of these edits changes Workbench. Relabelling or confirming a subject alone must not restrict label-independent analytical checks to the chosen classes. Eligibility changes alter their sample denominator. A reported inability to recompute must be read before treating a displayed value as an updated result.

Confirmed subjects change the informational feature-association test, not class separability, the class-count condition or the four analytical scores. The count condition counts sample IDs, not independent subjects. Grouped train/test or cross-validation splits remain a downstream responsibility; MERIT-ML does not construct or validate those folds.

6. Structural diagnostics

Structural has **four scored items** and a separate diagnostic mean. It does not set analytical alerts or the count-support result. Values below 100 identify the relevant issue; 100 does not rule out all structural problems or subject dependence.

6.1 Sample support

Score = `100 × min(1, n / N)`, with `n` distinct ML-eligible sample IDs and `N = 20` by default. N can be 5–100 in steps of 1. With no eligible sample it is not assessed. The target is a descriptive working reference, not a power calculation, per-class threshold or readiness gate. Changing N affects this item and the Structural mean, not the k-fold count requirement.

6.2 Samples with a factor label

Score = `100 × eligible positions with a non-empty factor label / eligible positions`. A label is non-empty when at least one value is filled. Only an empty value or `-` is empty for this check; deposited words such as `None` and `NA` remain values. A label can describe batch, age or an irrelevant factor and still count. It need not be an appropriate prediction target or have every constituent factor filled. No eligible positions means not assessed.

6.3 Numeric value cells

Score = `100 × numeric cells / (numeric cells + text cells)` over eligible positions. Recognised missing cells are excluded from this denominator. Empty cells and missing tokens such as NA, NaN, null, `-` and `.` are missing; detection-limit entries, spreadsheet errors and malformed numbers can count as text. The parser recognises the supported malformed-exponent notation as numeric where specified. With no numeric or text cell remaining, the item is not assessed. This differs from the missing-entry rate, which counts all empty/non-numeric cells against all eligible value cells.

6.4 Duplicate sample IDs

The positive score is `100 × distinct eligible sample IDs / eligible positions`; 100 means no repeated ID among those positions. The tab is headed **Duplicate sample IDs**, while some tables describe the positive quantity as distinct sample IDs per position. IDs are not automatically treated as independent samples or as known technical replicates. Repeated measurements should be combined appropriately or kept on one side of a train/test split.

6.5 Structural mean and informational evidence

The displayed Structural mean averages only the assessable items. No assessable items means no mean, not zero. The number assessed matters. High means do not cancel specific defects.

Details retain deposited/measured/eligible positions and distinct-ID counts; repeated positions and IDs; feature/value-cell counts; missing/text/numeric counts and text classes; completeness of all factor values; factor-label provenance; sample-table membership and sample-table line counts; declared non-control samples represented in the matrix; subject coverage, distinct/shared subjects and dependence warnings; Tier 1 alignment; and agreement of sample identities/position counts and feature counts across endpoints. Endpoints can share IDs yet have different numbers of positions, one can be a subset, they can partly overlap, or they can have none in common. A count difference alone does not identify the cause; inspect the IDs and source evidence.

7. Metadata and FAIR Reusability

This section has **five scored components**, assessed per analysis except study metadata, which is shared. It reports a separate documentation index, not a modelling certificate or formal comprehensive FAIR certification. Higher values mean more of the specified documentation or linkage is present.

7.1 Study metadata

Seven yes/no checks cover a DOI, a linked publication, funding, contributors, declared project/study type, a project or study summary of at least **20 words**, and documented raw data. Score = `100 × checks passed / 7`. Raw-data evidence can come from the study page or a data-format/raw-file declaration; it is documentation, not verification that the raw files are complete or reprocessable. Expand each item for the deposited evidence and source.

7.2 RefMet mapping of deposited metabolite records

Score = `100 × records with a valid deposited RefMet name / metabolite records` for the analysis. Records present but no valid mapping yield zero; no records means not assessed. Blank, `-`, Unknown, Standard and Protein placeholders are not valid RefMet names. This is a **record-level** component: duplicates remain records. MERIT-ML uses Workbench's assigned mapping; it does not independently identify a metabolite.

Do not confuse this with the Annotation tab's mapping of **named matrix features**, which has another denominator and can differ by endpoint.

7.3 External database cross-references via RefMet

Score = the percentage of valid RefMet-mapped records whose RefMet term has at least one cross-reference in the implemented ChEBI, InChIKey, PubChem, HMDB, KEGG or LIPID MAPS fields. No valid RefMet records means not assessed. These are links supplied through RefMet, not necessarily identifiers deposited by the study. Having one does not prove that the experimental annotation or structure assignment is correct. A mapped lipid/class term can lack a structure-level identifier.

7.4 Mass/RT-like metadata presence

This binary component is 100 when the mwTab annotation evidence contains a filled mass/m/z-like or retention-time/index-like field under the implemented rules, otherwise zero. An mwTab record without an annotation table scores zero; a record that cannot be read is distinguished from absent coordinate content.

The scored evidence concerns annotation metadata for named metabolites. The m/z–RT text of an unnamed feature or an untargeted results-file header is displayed as contextual evidence and is not silently counted as this annotation field. Presence of exact/neutral mass is not the same as a measured m/z, and retention index is not retention time.

Coordinate details show field provenance and coverage, mass-like versus strict m/z-like evidence, RT/RI evidence, coverage of either/both coordinates, RT-unit category and adduct declarations where present. These explain the component; they are not extra scored metrics or proof of identification confidence.

7.5 Basic LC-method context

Only analyses recognised as explicitly LC by the implemented chromatography-type rule are assessed. Score = one third each for a column declaration, **both** mobile phases and a gradient/composition declaration. The gradient rule requires percentage/composition wording with time information or an isocratic declaration. Dedicated chromatography fields are read; an external method file or unstructured narrative is not assumed to supply missing components.

Flow rate, temperature, dimensions, particle size and basic/extended/full-context details are informational. GC or another explicit non-LC type is not applicable. An unknown/unrecognised type is shown with the actual declaration or “LC not explicitly identified”, rather than proof of no LC. LC context does not demonstrate comparable retention times across studies.

7.6 FAIR documentation index

The index is the arithmetic mean of **assessable** FAIR components, so different analyses can have different numbers of components in the denominator. Unavailable components are not automatically zero; a measured absence in an assessable binary component is zero. Inspect the component values and number assessed. The index never offsets analytical alerts, changes count support or contributes to a universal readiness score. Deposited values, lookup provenance, coordinate details and LC components remain available alongside it.

8. Analytical QC: samples, controls and processing

Analytical QC contains **five scored items**: Missing entries and zeros, Low-information features, Redundant features, Outlier samples and Units declared. Only the first four can raise analytical review alerts. The remaining blocks below are informational, including QC reproducibility.

8.1 Sample eligibility and analytical controls

The panel counts measured positions, eligible positions, QC-like, blank-like and standard-like positions, Tier 1 alignment, and edited exclusions/set-aside positions per matrix. It also distinguishes controls declared in the sample table from controls actually measured in this endpoint, including **Not measured here**. Sample-table counts and measured-position counts are different quantities.

The default keyword rule searches sample IDs and **factor values**, not factor names. A keyword in the sample ID decides before a keyword in factor values; blank matches take precedence over QC matches. The curated correction can override a keyword or supply a documented additional QC, blank or standard. Every position with that sample ID follows its per-analysis identity rule; a sample can have a different alignment/measurement status in another matrix. Details give keyword, location, source label and curated evidence. The QC keyword set is `qc`, `nist`, `ltr`, `sst`, `pool`, `reference`, `quality control`, `pooled qc`, `system suitability`, `calibration standard` and `drift`. The blank set is `blank`, `solvent`, `process blank`, `method blank` and `reagent blank`. Matching is case-insensitive; qc/nist/ltr/sst do not match immediately after a letter. Names alone do not verify material or injection type. Full lists are downloadable; a short on-page example list need not display every ID.

8.2 QC reproducibility: median QC RSD

This is **informational**, with lower RSD indicating less relative spread in repeated measurements. It does not raise an alert and is not averaged into a quality score.

QC-like positions are grouped by deposited material label, or by sample ID when that label is empty. The **largest group** is used, requiring at least **three injections** of one QC material. Blank-like and standard-like positions are not substituted. Units must indicate a supported linear family—concentration/amount, intensity/area/counts, or normalised/relative/ratio—and negative values must be absent. Log-indicated, other/mixed or undeclared units, insufficient injections or unavailable values produce an explicit not-assessed reason.

For each feature, detected means a finite **non-zero** value. Detection percentage = detected injections / injections × 100. RSD = `100 × sample SD / mean` over the detected values, with SD using `ddof = 1` and at least two detected values. The headline median RSD uses only features detected in **more than 70%** of injections.

A passing feature has detection >70% **and** RSD strictly below the selected cut-off, default **20%**, adjustable **10–40%** in steps of 1. Exactly 70% detection or an RSD equal to the cut-off does not pass. Changing the RSD cut-off changes passing/failing counts, not the headline median calculation. Details/downloads give material, injections, detected coverage, per-feature RSD and detection/RSD failure counts. Repeated IDs alone do not establish that they are the appropriate QC replicates.

8.3 Processing evidence

Read the quoted dedicated software-version, processing-parameter-file, analysis-detail and NMR-processing fields, software mentions and minFraction/minfrac wording. Details show named processing steps, where the words were found and record-identity warnings. SAMPLEPREP:PROCESSING_METHOD is not treated as computational processing evidence. A software mention is not proof it produced this matrix; a named parameter file is not parsed as though its contents were available. These are deposited evidence, not a verified preprocessing pipeline or normalisation score.

9. Analytical QC: the four alerted metrics

The four scores describe different properties and denominators. Their default R1 alert boundaries are given in section 15. Their settings change metric calculations; alert percentiles/custom cut-offs change how those values are flagged. Classes selected for a prediction task do **not** filter the samples used by these four label-independent checks.

9.1 Missing entries and zeros

The denominator is **all feature positions × ML-eligible sample positions**, including unnamed features. Each category's score is `100 × (1 − category cells / eligible cells)`. The displayed metric is the **lowest applicable category score**, equivalently 100 minus the largest category percentage. It is not 100 minus the sum of categories and is not an average of per-sample completeness scores.

Category	Meaning
I1: Missing entries	Empty or non-numeric cells, with the six entry classes reported separately. The current implementation calculates this category for assessed matrices, including datatable.
I2: Zeros, meaning unknown	Every mwTab/untarg_data zero. For datatable: zeros also zero in mwTab, not traceable to a paired cell, or in an unpaired datatable.
I3: Zeros hiding a missing value	A paired datatable zero whose matching mwTab cell has no number: empty, missing token or text.
I4: Zeros replacing a value	A paired datatable zero whose matching mwTab cell has a non-zero number, including negative/positive values and values that round to zero at datatable precision.

I3 and I4 apply only when paired evidence exists. All categories use the same eligible-cell denominator. A zero is not automatically called missing or incorrect. The metric reports prevalence and provenance; it does not change the deposited values.

Datatable zeros are traced by sample ID and **normalised feature name**, not row/column order. Feature normalisation trims, lowercases, collapses spaces, reads commas as underscores and apostrophes as carets. For repeated feature names or sample IDs, all candidate paired cells are considered and a category is assigned only when they agree; unresolved candidates remain untraced. Details identify extensions using repeated IDs/names. This differs from analyses restricted to unambiguous one-to-one cells.

The six missing/non-numeric classes are empty, missing-value token, detection-limit entry, non-finite number, malformed number and other text. Details include whole-deposited-matrix rates, eligible rates, per-sample distributions, samples without numeric values/all zeros, zero provenance and negative values written as zero. Whole-matrix figures include a different denominator from the score. At zero eligible samples or zero features, the metric is not assessed.

Higher = a smaller largest category. A score of 100 means no cell falls into a scored category, not verified absence of every analytical problem. A negative value written as zero has its own provenance flag; numerical rounding and clipping cannot always be distinguished from the datatable alone. For modelling, review source-specific missingness and do not uniformly turn all zeros into missing values.

9.2 Low-information features

The input is numeric feature values on eligible **positions**; invalid/missing entries are omitted and zeros are real numeric values. A feature is **judged** with at least three numeric values. Constant means one distinct numeric value. For non-constant features, near-zero variance means both:

- count of most common numeric value / count of second most common value **> R**, default 19;
- `100 × distinct numeric values / eligible positions ≤ U`, default 10%.

R can be 5–99 in steps of 1; U can be 1–25% in steps of 1. Exactly R does not meet the frequency rule; exactly U does meet the uniqueness rule. Missing positions remain in the uniqueness denominator. The score is `100 × (1 − (constant`

`+ near-zero-variance features) / judged features)`. Each feature is counted once. Fewer than five eligible positions or no judged feature makes it not assessed. Features with too few values are reported rather than assumed informative.

Higher = fewer judged features flagged by this rule. **Examples** shows up to five flagged features, constants first. Details distinguish constant, near-zero variance and unjudged features, with observations and active settings. The full JSON list is generated on request for every relevant analysis × endpoint, with names/reasons and settings, rather than embedded as every feature list in the report.

9.3 Relative-spread caution (CV)

This belongs beside Low-information features but is **not scored**. Among features not already flagged, CV = `100 × sample SD / mean`, with at least three numeric values, positive mean and SD `ddof = 1`. It is considered only for declared concentration/amount or intensity/area/count units with no negative values. It is not interpreted on log-indicated, normalised, other/mixed or undeclared units.

The caution counts features with CV **below** C, default 1%, adjustable 0.1–5% in steps of 0.1. These can be internal standards, rounded/fixed quantities or genuinely stable metabolites. The value is not proof of a defect. Changing C does not change the low-information score or analytical alert; the full low-information download also includes the caution features.

9.4 Redundant features

Features must have at least three values and not be flagged by the current low-information rule. Each pair requires at least **10 shared numeric positions**. Missing entries are omitted; zeros remain values. The metric needs at least **10 eligible positions** and at least two evaluated features, otherwise it is not assessed.

The implementation ranks each feature once over its observed values, using average ranks for ties. It correlates those ranks on the pair's shared observations. With complete data this is usual Spearman correlation; with missing data it need not equal separately re-ranking every pair's shared subset. Absolute correlation is used, so strong negative and positive relationships can both count.

The threshold c defaults to **0.90**, adjustable **0.80–0.99** in steps of 0.01. A pair must exceed c; the implementation treats values within `1e−5` of c as equal, to avoid numerical rounding artefacts.

The **fast caret-style selection** flags, for each exceeding pair, the member with the larger mean absolute correlation across the evaluated features; its documented tie rule resolves equal means. A feature selected by multiple pairs is counted once. Mean correlations are not iteratively recalculated after each removal. Score = `100 × (1 − flagged features / evaluated features)`. Removing the flagged set leaves no evaluated pair above the threshold, but the set is not guaranteed to be the smallest possible removal set.

Higher = fewer features selected for removal by this rule. The count of correlated **pairs**, count of **flagged features**, and largest **connected group** are different. Connected does not mean every member is correlated above the threshold with every other member. Examples show up to five strongest pairs; downloads include flagged features with strongest partners and the exceeding pairs for all analyses/endpoints. A small-sample caution is shown at 10–19 eligible samples; this does not alter the score. Changing low-information settings can change which features are evaluated for redundancy. Changing labels/subject confirmation alone should not recalculate a different label-independent result.

Redundancy is method-dependent: correlated ions, isotopes or biochemical signals need not prevent predictive modelling. Review dimensionality, model choice, interpretation and feature selection rather than treating an alert as a universal exclusion.

9.5 Outlier samples: sample-median outliers

For each eligible position, take the median of its numeric feature values, **including zeros**. A position with no numeric median is excluded from this diagnostic's denominator and counted separately. At least five positions with a median are required.

Let Q1 and Q3 be the quartiles of the position medians and IQR = Q3 − Q1. A position is flagged when its median is **below** `Q1 − a × IQR` or **above** `Q3 + a × IQR`. The multiplier a is **1.5** by default, adjustable **1.0–3.0** in steps of 0.1. Values equal to a fence are not outliers. With IQR = 0 the fences collapse; the report states this rather than assuming no outlier is possible.

Score = `100 × (1 − flagged positions / positions with a numeric median)`. Higher = fewer medians outside the selected fences. Details give evaluated positions, missing medians, quartiles, IQR, fences, IDs, medians, side of fence and examples; a full list is downloadable. A repeated ID can occupy several tested positions, so an ID alone is not necessarily one unique observation. The matrix export's position field helps locate the deposited row/column.

The test concerns **sample medians only**. A biologically unusual multivariate feature profile can pass; zeros or sample-amount differences can change medians. Do not automatically discard every flagged sample or interpret 100 as no possible outlier.

10. Analytical QC: scale, units and contextual evidence

10.1 Units declared

This is Analytical QC's fifth scored item: **100** for a valid declaration and **0** otherwise. It is a documentation note, not one of the four alerts. A valid declaration is non-empty, non-placeholder text; NA, not applicable, `-` or a number alone do not supply usable units.

For mwTab/datatable, the quantitative data-block UNITS declaration is used for that analysis. Datatable shares this declaration; it is not independently verified by datatable numbers. For untarg_data, the declaration associated with the mwTab results-file line is considered separately. A data-block declaration is not automatically assigned to untarg_data. Missing declaration evidence is not proof that measurements have no physical units.

10.2 Deposited measurement scale and unit family

Scale descriptors are informational: the median and quartiles of non-zero numeric values, including negatives; negative-value counts/share; positive-value 5th–95th percentiles and orders of magnitude; and declared unit family. Values are not transformed, converted into common units or compared as if every endpoint used the same measurement scale.

Families include concentration/amount, intensity/area/counts, normalised/relative/ratio, log indicated and other/mixed, plus no declaration. The family comes from wording and rules, not a guarantee of the preprocessing history. It controls whether the optional CV/RSD calculations are meaningful under their implemented assumptions.

For eligible samples with positive medians, the panel can show their median, the 5th–95th percentile spread in orders of magnitude and samples more than tenfold from the typical sample median. Negative/zero medians are not silently put on a logarithmic axis. Paired datatable evidence can show whether non-zero mwTab numbers are retained to deposited precision. None of these descriptors has a universally better high or low value.

10.3 Platform/acquisition and per-analysis data summary

The platform/acquisition block reports analytical platform, analyzer capability, relevant wording, scan modes, repository/file markers and linked MS2 evidence with source attribution. The same evidence can also appear under Derived context on Overview. It is informational and does not verify targeted/untargeted acquisition or annotation confidence.

Per-Analysis Data Summary brings together available matrices, features, measured/eligible samples, controls, feature-to-sample ratio, annotation and numerical context for each analysis/endpoint. These quantities keep their original denominators; they are not an additional score. Data availability, context and documentation can be useful even when class counts or an analytical check are unavailable.

10.4 Sample annotations: MetaBatch-style filters

This informational analysis reads technical annotations from factor records, additional sample metadata and the batch-like edit. It recognises batch, plate, run order and run/day-type fields by names and, for batch/plate/run-like labels, their values. A candidate must have at least two distinct values, non-missing values for **at least 60%** of eligible positions, and distinct values no more than **90%** of positions. These filters are fixed, not Show tools sliders.

Details state the field, provenance, coverage, distinct values, whether it passes and why it fails. IDs, control-related fields and constant/nearly unique fields are not automatically suitable batch labels. A reported factor association can flag that each batch level contains one factor value, including a one-to-one or more general nesting relationship. This is evidence of a possible confounded design, not a fitted correction, a batch-effect test or proof of absence of batch effects when no field is found. It does not enter the readiness profile or FAIR index.

10.5 Additional sample data

Other sample-table fields are shown with field name, number of sample-table **records** reporting a value, distinct-value count and examples. Repeated sample-table lines count as records, not independent samples. The source can be

mwTab text or its JSON fallback and is identified.

Fields already represented as raw-file evidence, factors, identical factor values, sample IDs, subject IDs or technical sample annotations are not repeated; exclusions and reasons are available. A field name alone does not establish a phenotype, diagnosis or verified covariate.

11. Annotation / Interoperability

There are **two scored items**, with name/mapping/class/repetition evidence. They are separate interpretation/reuse indicators; unnamed features can still support modelling and naming alone does not establish correct metabolite identification.

11.1 Named-feature coverage

Score = `100 × named matrix features / eligible feature labels`. Feature labels are processed by the priority-ordered classifier. Named features include supported named/name-resolved categories. Features without a reported metabolite name include unresolved identifiers, class-level/tentative labels, formulae, m/z–RT, m/z-only, NMR bins, generic feature IDs and unknown labels. Empty/artifact/sample-metadata/protein labels and recognised malformed matrices are excluded according to the implemented rules.

The score is not assessed if no eligible feature remains. Counts of every category and exclusions, with examples, remain visible or downloadable. This annotation denominator differs from all deposited feature positions used by cell-based QC. Sample eligibility or class edits do not rename features or change the annotation calculation.

11.2 Named features mapped to RefMet

Score = `100 × named matrix features with a valid RefMet match / named matrix features`. Feature names are matched to metabolite records from the **same analysis**, after trimming, lowercasing, collapsing spaces, comma-to-underscore and apostrophe-to-caret normalisation.

Blank and invalid RefMet names are discarded first. Exactly one distinct valid RefMet name among matching records means mapped; several different valid names mean conflicting/unmapped. No named matrix feature means not assessed. MERIT-ML does not add a new chemical identification. This feature-based score and FAIR's record-based mapping can differ without being contradictory.

Details show mapped/unmapped/conflicting counts, reasons and examples, distinct mapped RefMet names, shared names and feature-level lists. Several ions, retention-time variants or features can map to one RefMet term; the distinct-name count need not equal the mapped-feature count.

11.3 Metabolite classes and repeated feature names

The metabolite-class block summarises RefMet superclasses; main/subclass information and per-feature assignments are in downloads. Classification coverage is limited by the available mapping/reference data, and a superclass is not a confirmed pathway result.

Repeated feature labels are grouped using normalised label keys; supported duplicate suffixes such as `.1` or `(2)` are ignored for named features. Otherwise an RT, ion or isotope suffix can distinguish labels. The block also identifies different labels sharing one RefMet name. Neither a repeated label nor a shared RefMet term proves an accidental duplicate analyte. Examples, group sizes and full JSON lists are provided. These counts do not enter either annotation score.

12. Label Structure and Class Support metrics

12.1 Class support

Let m be the smallest selected class, M the largest, and k the selected number of CV folds. Set `p = m / (m + M)` and `H(p) = −p log2(p) − (1 − p) log2(1 − p)`. The score is:

`Class support = 100 × H(p) × min(1, m / k)`.

The entropy term describes balance of the **smallest against largest** class, not entropy across all classes. The second term describes how far the smallest class reaches the fold count. At default k = 5, 10 versus 10 gives 100, 3 versus 3 gives 60, and 100 versus 200 gives about 91.8. One class gives zero. No labelled eligible samples or a selected class missing from this matrix makes it unavailable.

Higher means stronger support under this explicit heuristic; it is not a validated prediction of accuracy, power or estimate stability. It does not enter the analytical alerts. Class details show the balance and fold terms separately, samples left out and the factor/joint-class tables.

12.2 Cross-validation sample-count support

The tab can label this binary scored item **Cross-validation feasible**. Its precise meaning is the **count requirement**: 100/met when there are at least two classes and every class contains at least k distinct labelled eligible sample IDs; otherwise 0/not met. k defaults to **5**, adjustable **2–10** in steps of 1. Exactly k meets the condition. If the selected classes cannot be assessed, it is unavailable instead of 0.

This checks whether each class can appear in every stratified test fold. It does not execute cross-validation, train a model, certify independent subjects, assess statistical power or guarantee stable estimates. Five samples per class can meet the minimum yet be insufficient for a useful scientific model. Grouped subject-aware validation can have different requirements.

Changing k affects the count condition, the fold term of Class support, the profile and dependent guidance. It does **not** change the separability-neighbour setting, raw analytical scores, R1 reference or annotation/FAIR values.

13. ML Task Readiness metrics

These results use the selected classes. Separability is the section's one scored item, displayed with its chance level. Neither separability nor feature associations determine analytical alerts or replace validation of a model.

13.1 Class separability

Features are internally converted to average ranks across measured positions, with missing entries assigned the feature's mean rank, then centred and scaled per feature. Constant/no-variation features contribute no rank separation. Repeated positions of a sample ID are averaged into one profile. Distances are Euclidean on these profiles. This internal representation supports unit-agnostic comparisons; it is not an imputation/normalisation recommendation or a transformation of the exported deposited matrix.

Let K be the **number of nearest neighbours** (default 7) and θ the overlap threshold (default 3). For each class c, R(c) is the share of its samples with **more than θ** of the K neighbours belonging to other classes. Score = `100 × mean over classes of (1 − R(c))`. Every class gets equal weight. Higher means stronger local separation under the current labels; it is not cross-validated model accuracy.

K can be **3–15** in steps of 1. θ is a whole number, at least zero and at most **floor(K/2)**. The displayed input range extends to 7; its effective upper bound follows K and larger entries are capped. Distance ties at the K-th neighbour are handled by the implemented exact tied-neighbour calculation; a relative tolerance of `1e−5` accounts for stored precision.

The score is not assessed with fewer than two classes, an absent selected class, no varying usable feature, at most K samples in total or a class with fewer than **K − θ + 1** samples. The default K=7, θ=3 therefore needs at least five samples in each class, but this is a separability requirement distinct from the CV-fold control. For matrices above 4,000 sample IDs the retained neighbourhood has 200 neighbours per sample; selecting a narrow task can leave too few retained neighbours for an exact result, which is reported.

The rank representation is constructed for the matrix; class selection and eligibility changes select profiles rather than rebuilding a new rank transformation for each task. Consequently this is a diagnostic on the assessed matrix, not a leakage-safe feature-transform pipeline for estimating predictive performance.

13.2 Chance level and overlap details

Chance is the expected score if class labels were randomly assigned while preserving class sizes. For a class with n_c of N samples, the other-class-neighbour count follows the implemented hypergeometric distribution with population N−1, other-class population N−n_c and K draws. Chance = `100 × mean over classes P(other-class neighbours ≤ θ)`.

Always read separability beside chance. Strong imbalance and different class counts can change the chance value. The report can show class-specific overlap and chance, the most overlapping class pair, samples surrounded by other-class neighbours and tie evidence. These are exploratory label/data diagnostics, not evidence that samples are mislabeled or that a causal biological difference exists.

13.3 Features associated with selected classes

This informational block tests whether measured feature values differ between the selected classes. It reports the number/share tested and significant, test name, adjusted and raw p-values, direction where meaningful, class medians/counts, examples and reasons features/tests are unavailable. It is not automatic feature selection, model feature importance, a readiness score or proof of a disease biomarker.

Without confirmed subject information, the default is **Kruskal–Wallis**, using one value/profile per sample ID and testing a feature only when every selected class has at least **five measured values**. Missing values are excluded for that feature. These statistical tests use the implementation's rank-based representation; class medians/directions are presented from deposited values for the observations included by that feature's test.

With real subject IDs confirmed:

- When no subject spans classes, a supported independent comparison uses one value per subject; same-subject repeats do not count as new independent subjects.
- For two classes with shared subjects, the supported paired comparison is the **exact Wilcoxon signed-rank test** on complete subject pairs. Zero differences are omitted. At default q=0.05, at least **six non-zero pairs** are needed; the minimum follows `ceil(1 + log2(1/q))`.
- For more than two repeated classes, the supported **Friedman** test requires at least **11 complete subjects and seven classes**. Unsupported or insufficient designs are not assessed with subject adjustment.

For each feature, displayed medians and counts use exactly the subjects/observations retained by that feature's test, not every subject in the original table. Partial subject coverage, no complete pairs/blocks, too few non-zero differences, constant data or unsupported repeated designs have explicit reasons. Confirmation can reduce the assessable set rather than improve a score.

P-values for the tested features are adjusted by **Benjamini–Hochberg**. q defaults to **0.05**, adjustable **0.01–0.20** in steps of 0.01. Changing q changes the significance criterion and can change paired-test minimum support; it does not change class separability or the analytical alerts. Interpret two-class direction from the displayed class names/medians; a multiclass association need not have one meaningful up/down direction. Examples are limited; the strongest stored/available results and downloadable evidence should not be mistaken for every possible biomarker.

13.4 Features per sample and subject information

The task p/n uses the matrix's feature count and distinct eligible sample IDs in the selected classes; a zero selected-sample denominator is unavailable, not infinity used as advice. p/n elsewhere can use a whole-matrix or different sample denominator. High p/n raises overfitting and feature-selection considerations, not an automatic failure threshold.

Subject information reports coverage, distinct subjects, subjects with several samples and subjects spanning classes. Deposited unconfirmed IDs trigger verification advice; confirmed sharing calls for subject-disjoint validation. A single confirmed subject cannot support testing generalisation to unseen subjects. No subject information is not proof of independence. These warnings are separate from the numerical sample-count condition.

14. All numerical controls

Open **Show tools** to see the current settings and Help. Apply a change to recompute the relevant outputs. The CV-fold control is prominent because it directly changes count support; it is a different k from separability neighbours. Settings shared between report sections are one active choice for the study.

The following table is generated from the current calculation registry and checked against the rendered inputs. **Default / minimum / maximum / step** describe the actual numerical controls; read these inputs and exported settings rather than rounding in small display badges.

Setting	Default	Minimum–maximum	Step	Effect / constraint
Cross-validation folds k	5	2–10	1	Class-count requirement and Class support fold term.
Structural: sample-support target N	20	5–100	1	Structural item and mean; not CV count support.
Low-information: frequency ratio R	19	5–99	1	Flag when ratio > R and distinct percentage ≤ U; also changes redundancy eligibility.
Low-information: distinct-value percentage U	10	1–25	1	Flag when percentage ≤ U and ratio > R; also changes redundancy eligibility.
Low-variation caution: CV (%)	1	0.1–5	0.1	Informational only; strictly below the threshold.
Redundancy: absolute correlation c	0.9	0.8–0.99	0.01	Evaluated pairs above c, with 1e−5 equality tolerance.
QC reproducibility: RSD (%)	20	10–40	1	Informational pass counts; detection must still be >70%.
Outliers: fence multiplier a	1.5	1–3	0.1	Sample medians outside Q1−a·IQR / Q3+a·IQR.
Extreme-alert percentile	2.5	2.5–10	2.5	Choose 2.5, 5, 7.5 or 10 from R1.
Low-alert percentile	10	2.5–10	2.5	Choose 2.5, 5, 7.5 or 10; effective value never below extreme.
Separability: neighbours K	7	3–15	1	Neighbour count, not CV folds.
Separability: overlap threshold θ	3	0–7	1	Integer; effective maximum floor(K/2), although input maximum is 7.
Feature associations: FDR q	0.05	0.01–0.2	0.01	Significance and paired-test minimum support.

The θ cap and permitted alert percentiles in the notes are additional constraints. There are **13 registered numerical settings plus eight optional custom cut-off fields**, not 21 independent scored metrics. All other fixed requirements, such as detection >70%, at least three values, minimum shared-pair samples and MetaBatch-style coverage, are described with their metrics rather than exposed as sliders.

14.1 Custom alert score cut-offs

For **each of four checks**, enter an optional **extreme** and/or **low** score cut-off: eight fields. Allowed numbers are **0–100**, including decimals. Empty means use the selected R1 percentile. Non-numeric, non-finite or out-of-range entries are ignored in favour of the reference and their ignored status is reported. An alert requires the score to be **strictly below** its cut-off.

If the low cut-off is below the extreme cut-off, the effective low cut-off is raised to the extreme one; low cannot be narrower than extreme. A custom cut-off is identified as custom where relevant and is retained in the assessment settings. An extreme cut-off of zero cannot flag a non-negative score. A cut-off of 100 can flag every assessable score below 100, not scores tied at 100.

Changing these values changes alert assignments and guidance, not raw metric scores. It does not create a new reference cohort or rewrite R1. Restoring the defaults clears the custom fields. The two shared editors use the applied values in the report and exports.

14.2 What each change can affect

Change	Outputs that depend on it	Outputs not recalibrated by that change
CV folds	Count support; Class support fold term; profile; related guidance	Analytical scores, R1, separability neighbours, annotation/FAIR
Selected factors/classes	Class counts/support; separability/chance; feature tests; selected-task p/n; other-factor comparison; task narrative	Label-independent analytical checks and annotation/FAIR
Sample eligibility/exclusion	Applicable Structural/analytical metrics; class metrics; selected-task outputs and guidance	Deposited source values and reference R1
Factor-label edits	Assignment and factor selection; related control/eligibility evidence where the rules use it; class-dependent outputs	A pure label change is not permission to filter analytical QC to selected classes
Subject edits/confirmation	Subject evidence and supported subject-aware feature tests	Count support, separability and analytical scores
Batch-like edit	Sample-annotation fields and associated informational warnings	Analytical scores, class labels and FAIR index
Low-information R/U	Flagged features and score; evaluated redundancy features; alerts and details	Reference cut-offs themselves
CV caution C	Informational low-variation count/list	Low-information score and alerts
Redundancy c	Exceeding pairs, selected features, redundancy score and related alerts	Other analytical scores
Outlier multiplier	Fences, flagged sample medians, outlier score and related alerts	Other analytical scores
QC RSD cut-off	Per-feature QC pass/fail counts	Headline median RSD and analytical alerts
Separability K/θ	Separability, chance and overlap details	CV fold count and feature-test q
Feature-test q	Adjusted-significance summaries and q-dependent paired support	Separability and analytical alerts
Alert percentiles/custom cut-offs	Alert tiers, counts, summary and corresponding narrative	Metric values, classes, R1 source distributions
Theme/tour/Show numbers/details	Presentation	All scientific inputs and results

An unavailable/resource-limited recomputation remains unavailable with a reason. It must not be inferred to have succeeded because a control accepted a value. Save the assessment record to document the effective settings and omissions.

15. R1 and alert interpretation

R1 is a **versioned reference of default analytical scores**, originally calculated from 11,749 valid matrices in the 4,121-study manuscript benchmark. It is not an overall score, experimental-quality ranking or threshold learned from downstream accuracy. The reference counts matrices, so a study with several analyses/endpoints can contribute several observations.

At defaults, an assessable score strictly below its **2.5th-percentile** cut-off is extreme. A score below its **10th-percentile** cut-off but not extreme is low. Exact equality is not alerted at that boundary. Missing/unassessed scores do not enter the percentile calculation or become passes. Ties mean the share flagged need not equal the nominal percentile.

Percentile	Missing entries and zeros	Low-information features	Redundant features	Outlier samples
2.5 (default extreme)	30.30	68.91	21.38	80.00
5	42.22	87.50	28.57	83.33
7.5	50.00	93.60	34.19	86.67
10 (default low)	57.01	96.97	40.00	88.24

Reference populations with an assessable value: Missing entries and zeros: **11,717**; Low-information features: **11,518**; Redundant features: **10,367**; Outlier samples: **11,591**.

The table shows rounded cut-offs; full precision is used in calculation. Each percentile uses only matrices where that particular metric could be assessed. The chosen default low and extreme percentiles can each be **2.5, 5, 7.5 or 10**, with low never below extreme. Selecting another percentile changes which reference cut-off is used; it does not recalculate the reference from the current study.

The low boundary controls the total set reviewed; moving only the extreme boundary redistributes flagged matrices between extreme and low tiers. Several checks can flag one matrix, so their alert counts cannot be added to obtain unique matrices. The summary uses the strongest tier and shows how many checks were assessed.

R1 remains the same when a new study is queried. Entering custom values overrides the decision boundaries, not the reference's underlying cohort. Future recalibration needs a documented new reference version, with its cohort and calculation rules preserved. The assessment record identifies the reference and active custom settings. A planned maintenance audit is not a current per-request automatic recalibration feature.

The reference describes all assessed Workbench matrices, not experimentally matched peers. Endpoint, platform, sample size, preprocessing and feature composition can affect these distributions. An extreme alert is a prompt for specific review, not a declaration that the matrix is unusable. No alert means no assessed value crossed these relative boundaries; it is not certification.

16. Downloads and provenance

16.1 Assessment JSON

Use **Download MERIT-ML Assessment JSON** at the report top. It records current per-analysis/endpoint results, supporting counts, applied parameters, class/sample edits, subject confirmation and reference/custom cut-off information. No universal readiness score or old band should be inferred from this profile. Save records when comparing tasks or settings; the on-screen page is not a publication archive.

16.2 Matrix with classes in force

Data at a glance offers an endpoint-named **matrix CSV**. It has one row per ML-eligible **position**, with sample ID, the original position index (starting at 1 among matrix sample positions), subject ID, class in force, deposited factor string and one column per feature. Repeated sample IDs can therefore produce several rows. The exported subject reflects the supported effective subject handling; retain the accompanying assessment record when documenting edits.

All eligible rows are retained, including rows **outside the selected classes**, whose class field is empty. Selecting two classes does not automatically drop every other eligible row from this export. The caption explains this; filter non-selected rows explicitly for a selected-task model. QC/blank/standard-like positions, alignment positions and explicitly excluded samples are not eligible matrix rows.

Numeric value cells remain as deposited; this is not imputation, scaling, normalisation, feature removal or subject aggregation. Do not feed duplicate positions or unassigned rows blindly into a model. The download name describes its matrix/classes rather than certifying it as model-ready.

16.3 Deposited-position matrix ZIP and source inputs

The per-matrix input download can preserve every deposited position with status/eligibility and feature cells, together with a manifest. This is distinct from the eligible-row CSV. Use the export's column definitions and manifest to distinguish full-position data from modelling-subset data. Underlying source downloads preserve endpoint identities; datatable, mwTab and untarg_data are not silently merged.

The browser's input/run-record downloads include the source bodies actually retained, checksums, retrieval information and omissions. A refused or interrupted body does not receive a fabricated full-file hash. A bounded metadata-only view is identified as derived, not presented as the complete raw response. A run record is not the same as the smaller assessment JSON.

16.4 Feature lists and other detailed downloads

The small **download the full list (JSON)** controls generate analysis × endpoint lists for the active inputs. Available kinds include controls, low-information/CV cautions, redundant features/pairs, outlier samples, QC reproducibility, annotation coverage/mapping, metabolite classes and repeated-name groups. Acquisition/context evidence has its own structured download. Lists show supporting reasons or categories rather than another overall score.

Lists are built when requested, avoiding the need to embed every large list in the page. Building a download still needs computation and memory; it does not remove the storage needed for the current study's underlying inputs and calculations. Large exports can fail their resource check while the assessment remains valid and visible.

16.5 Reproducibility and data handling

Record the study/analysis/endpoint, retrieval date and source hashes, software/reference version, effective settings, class choice, edits, omitted checks and exported data's scope. Workbench can change deposited records; a later fresh fetch can yield a different matrix. Report whether compared results came from the same fetched inputs. Downloaded source data remain attributable to the repository and study authors; a local export does not grant permission to redistribute those inputs.

17. Bulk assessment and local Docker use

17.1 Native bulk assessment

The landing-page bulk field accepts typed/pasted study IDs, normalises them and removes duplicates, with a maximum of **500 IDs** per submitted set. Review the validation results before running. Available IDs are green, invalid/unavailable IDs red, explicitly confirmed embargoes amber and unverified entries grey. A failed public-availability request is not proof of an embargo. Only available accepted IDs are submitted.

The completed bulk view gives summary tables/downloads and underlined **individual study-report links**. Open each in a new tab to inspect and interact with the ordinary report. The link retains the bulk job's settings/edits where applicable; later interaction in a study report does not silently rewrite the already completed bulk summary. Reports require retained job files; expired/deleted jobs have an explicit message.

The browser landing displays bulk assessment disabled/blurred with a link to Docker. It assesses one study per running session. The retired Find Similar Studies and old report-level bulk panels are not part of the current workflow.

17.2 Install the native application

Docker must be installed and running; Python/scientific dependencies are contained in the image. Follow the current repository instructions for the published release and platform. The supplied launcher requests an available **localhost** port; inspect the printed address rather than assuming port 8791 on your computer.

On Linux, prefix Docker commands with `sudo` if your account needs it. On Windows/macOS use the corresponding Docker terminal and omit `sudo` where it is unavailable. Docker provides the Linux container environment; this is not a claim that Docker itself requires no installation. The source build is an alternative to a published image; the repository gives exact image tags and Compose commands.

For this documented release, the published image can be started on an available localhost port:

```
 sudo docker run -d --name merit-ml --init -p 127.0.0.1::8791 -v merit-ml-studies:/app/runs
ghcr.io/biosystemengineeringlab-iitb/merit-ml:v2-live-20261112-r13
sudo docker port merit-ml 8791/tcp
```

Open the printed address with `http://` in front. On Windows/macOS, or an account already permitted to use Docker, omit `sudo` as appropriate. If a container with that name already exists, start the existing one or choose a new name rather than deleting data accidentally. `sudo docker stop merit-ml` stops it; `sudo docker start merit-ml` starts it again.

To build the source checkout instead:

```
 sudo docker compose -f compose.merit-v2.yml up -d --build
sudo docker compose -f compose.merit-v2.yml port merit 8791
```

Only studies you request are downloaded. A Docker volume retains local runs and can grow; there is no automatic whole-cohort download at startup or guaranteed 5 GB cache eviction policy. Export reports you need. Stopping a container differs from deleting a volume; removing the volume deletes retained local data. The native server normally expires completed web jobs after 24 hours unless configured to retain them. Internet access is required to retrieve repository inputs and external evidence.

17.3 Native resource settings

Environment setting	Default	Meaning
`MERIT_WORKER_MEMORY_MB`	Automatic, 65% of currently available host/container memory	Per-worker MiB budget; 0 chooses automatic. Leave memory for the server and other processes.
`MERIT_CHECK_TIMEOUT_SECONDS`	1800 seconds	Time limit for an individual calculation/check; 0 disables this timeout.
`MERIT_RENDER_TIMEOUT_SECONDS`	3600 seconds	Maximum wait for report/export rendering.
Docker memory allocation	Your container/host setting	Overall memory still bounds the application; increasing a per-worker setting does not create RAM.

These are deployment/resource controls, not scientific sliders. Execution records give budgets, elapsed time, measured peak RSS and error/timeout reasons. No fixed browser matrix-size ceiling is applied to the native application. However, finite memory/time, output construction, process failures or an OS-level termination can still prevent a calculation. The application is not a promise that any matrix size will finish on any computer. The default installation binds localhost; public exposure requires a separately secured service deployment.

18. Browser limits and partial results

The browser uses **Pyodide/WebAssembly** and JavaScript workers to run the assessment with your compute. Versioned application/reference assets are downloaded as needed. A study is not refused merely because one matrix is large: the guard considers each analysis × endpoint and, where applicable, each check. Successful results are retained and independent work continues.

The current admission estimate starts with **330 MiB** runtime and adds estimated costs for cells, input-file bytes, stored feature pairs and squared sample-position count. A bounded feature-pair probe can refine dense-pair estimates. The admission budget is **1 GiB**, or **0.5 GiB** when a device reports less than 4 GB memory. These are estimates, not measured universal peak-memory guarantees. Device memory can be unavailable or approximate.

Input responses are streamed with a **50 MB** limit; a declared oversize response can be refused before collection, and collection stops if the streamed body crosses the limit. ZIP, CSV/text and JSON output construction also has a **50 MB** browser limit, with additional pair-list preflight checks. A small report can therefore coexist with an unavailable large download.

When a memory/file restriction applies, a prominent notice names the affected endpoint/analysis and links to the full application. **Blank does not mean absent.** The report distinguishes absent data, failed transfers, refused matrices, missing shared metadata and checks that could not be assessed. Datatable zero provenance is not guessed when the necessary mwTab evidence was unavailable.

Saved checkpoints on the user's device retain completed tasks and omission/fetch provenance during a run. Worker replacement can recover from a failed calculation and continue remaining work. This does not guarantee recovery after the operating system kills the entire tab, browser storage is unavailable or disk quota is exhausted. Keep the tab open and export important completed results. Use Docker on a sufficiently resourced computer for a browser-refused matrix.

Eligibility/low-information changes can require expensive pair recomputation; a resource-limited interactive check is reported as unavailable. A pure class or subject edit does not by itself justify discarding a label-independent matrix. The main report and Data at a glance should identify incomplete work rather than reporting fabricated zero/pass values.

19. Troubleshooting and practical interpretation

What you see	What to do
Study rejected/unavailable	Check ST accession and public release; distinguish explicit embargo from an availability-request failure. Retry a network failure.
Metadata-only report	Inspect which endpoints lacked a readable quantitative block. Raw files/attachments are not automatically processed into a matrix.
No ML-eligible samples	Review controls, alignment and edits; no class/cell interpretation should be inferred from an empty eligible subset.
Cannot assess class counts	Check labelled eligible IDs and whether every selected class occurs in this particular matrix. Study-wide selector counts are not its matrix counts.
Count requirement not met	Review class sizes, scientific task and folds. A different grouping asks a different question; relabelling or synthetic oversampling does not supply new independent test subjects.
No alerts among fewer than four assessed checks	Read missing-check names/reasons; this is incomplete coverage, not all checks passing.
Extreme analytical alert	Read the named metric's absolute burden, denominator, settings and evidence. Handle missing/zero representation or appropriate feature/sample review before modelling.
Strong separability	Compare chance; review labels, subject dependence, other factors, batch and external validation. This is not measured test-set performance.
Shared/unverified subjects	Verify the identifier; if real, keep subjects disjoint in downstream validation. Do not assume “Human” or a group name identifies a person.
A chosen class has no samples after edits	Change the task or restore relevant eligibility only if justified; do not silently ignore the absent class.
Browser refusal or stopped worker	Keep completed results, read the named matrix/check and reason, and use the native application for omitted work.
Export fails but report remains	Check output/pair-list resource limits; calculation and export availability are separate.
Slow assessment	Allow for repository/network delays and feature-pair calculations; read current step and Generating report. Do not assume a quiet interval proves completion.

A practical review order is: verify source/matrix identity and Structural evidence; choose and justify classes; inspect eligible samples/subjects and count support; review analytical burdens and scale; then assess annotation/FAIR evidence for the interpretation or cross-study reuse you intend. Missing-value handling and feature selection for performance estimation belong **inside training folds**. Do not use MERIT's diagnostic feature associations as feature selection on the whole dataset before cross-validation.

MERIT-ML does not verify the complete acquisition/preprocessing pipeline, fit a predictive model, measure learning-curve stability, certify power, detect every mislabeled sample, rule out all confounding, confirm metabolite identities, guarantee privacy/fairness or validate independent-cohort performance. The profile is evidence for review under stated inputs and rules.

20. Complete metric and evidence checklist

The checklist below maps every registered reference-table item to this guide. Some items are summaries or informational evidence rather than additional numeric metrics. It also lists the newer interactive and informational features that were added after the earlier metric reference.

Item	Role	Guide section
1. Sample support	Scored	6
2. Samples with a factor label	Scored	6
3. Numeric value cells	Scored	6
4. Distinct sample IDs per position (tab: Duplicate sample IDs)	Scored	6
5. Structural mean	Separate summary	6
6. Linkage, completeness and agreement	Informational	6
7. Study metadata	Scored	7
8. RefMet mapping of deposited metabolite records	Scored	7
9. External IDs via RefMet	Scored	7
10. Mass/RT metadata	Scored (0/100)	7
11. LC-method context	Scored	7
12. FAIR index	Separate summary	7
13. Deposited evidence	Informational	7
14. Missing entries and zeros †	Scored; alert	8–10
15. Low-information features †	Scored; alert	8–10
16. Redundant features †	Scored; alert	8–10
17. Outlier samples †	Scored; alert	8–10
18. Units declared	Scored (0/100)	8–10
19. Analytical review alerts	Readiness profile	8–10
20. Sample eligibility and controls	Informational	8–10
21. QC reproducibility	Informational	8.2
22. Low-variation caution	Informational	8–10
23. Measurement scale	Informational	8–10
24. Acquisition, data summary and processing	Informational	8–10
25. Sample annotations (MetaBatch-style)	Informational	8–10
26. Additional sample data	Informational	8–10
27. Evidence behind the scored items	Informational	8–10
28. Named-feature coverage	Scored	11
29. RefMet mapping of named features	Scored	11

Item	Role	Guide section
30. Labels, mapping, classes and repeats	Informational	11
31. Class support	Scored	4, 12
32. Cross-validation sample-count support (tab: Cross-validation feasible)	Scored (0/100); readiness profile	4, 12
33. Classes, factors and exclusions	Informational	4, 12
34. Class separability	Scored	13
35. Chance level	Shown with the score	13
36. Features associated with the selected classes	Informational	13
37. Overlap details, p/n and subjects	Informational	13
Search, availability, progress and Generating report	Interface / informational	1, 3
Light/dark mode, About, publication, GitHub and manual	Interface / informational	1
Search/report tours, replay and opt-out/storage limits	Interface / informational	1.2
Per-matrix profile, coverage, tiles, colour/help and grouped endpoints	Interface / informational	3
Data at a glance and applied-input recommendations	Interface / informational	3.2, 19
Derived design/analyzer/acquisition context and provenance	Interface / informational	3.3, 10.3
Whole-string / selected-factor classes and observed combinations	Interface / informational	4
Other-factor states, proportions and joint distributions	Interface / informational	4.3
Use as, labels, batch-like edits, subject shortcuts and confirmation	Interface / informational	5
Thirteen numerical settings, bounds, steps and dependencies	Interface / informational	14
Eight custom score cut-offs, fallbacks and precedence	Interface / informational	14.1
Fixed R1, strict comparisons, ties and reference versioning	Interface / informational	15
Assessment JSON, matrices, full-position exports and feature/evidence lists	Interface / informational	16
Bulk validation/statuses, individual interactive reports and expiry	Interface / informational	17
Native memory/check/render settings and available-port launch	Interface / informational	17.2–17.3
Browser per-matrix guards, downloads, checkpoints and omissions	Interface / informational	18
Unavailable / partial / metadata-only states and troubleshooting	Interface / informational	2, 3, 18–19

21. Versions, provenance and further help

This manual was checked against the running report, its form controls, current calculation source and browser/native resource handling on **12 November 2026**. The accompanying coverage manifest records source-file hashes, the 13 numerical setting specifications, eight custom fields and metric/feature coverage. A documentation check does not establish that every future repository deposit will parse or that every matrix fits available hardware.

The current browser and native applications calculate matrix metrics from retrieved inputs. The manuscript's frozen benchmark is a separate reproducibility resource; the development reference viewer may display that benchmark rather than fetch a new study. All editions must identify their input and reference provenance. R1, chemical lookups, curated control rules and labelled contextual reference evidence are versioned inputs, not a claim that every contextual datum was freshly collected in the current assessment.

Save the assessment JSON and run record, where available, when asking for help. Include study/analysis/endpoint, active settings/classes, exact error, whether you used browser or Docker, and which independent results completed. Never send passwords, tokens, private keys or unrelated private sample information.

Visit MERIT-ML, the publication and the source, installation and issue tracker. This guide is available as searchable HTML, downloadable PDF and Markdown; its metric definitions and controls use the same manual version.

MERIT-ML user manual · 12 November 2026. Metric/control coverage and source hashes.